Mathematical and Computational Methods

Exercise Sheet 4: Vectors

Tutorial route (▶, in this order, about 75 minutes): 1.3(b,c), 2.2, 2.4, 2.7, 2.8, 2.9, 2.10. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill — component arithmetic and a sketch).* Let $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 2)$ in $\mathbb{R}^2$.

(a) Compute $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $3\mathbf{u}$ and $2\mathbf{u} - 3\mathbf{v}$.

(b) On one set of axes sketch $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$, each drawn from the origin, and check your first two answers to (a) against the sketch.

Exercise 1.2 *(⋆ drill — vector between points, distance).* For the points $A(2, -1, 3)$ and $B(5, 1, -1)$, find (a) the vector $\overrightarrow{AB}$ and (b) the distance from A to B.

Exercise 1.3 *(▶ ⋆ drill — unit vectors).* (a) Normalise $\mathbf{v} = (3, 4)$.

(b) Normalise $\mathbf{w} = (4, -4, 7)$.

(c) Find all unit vectors in $\mathbb{R}^2$ that make an angle of 45° with $(1, 0)$.

Exercise 1.4 *(⋆ drill — dot product, orthogonality and angle).* Let $\mathbf{a} = (2, 3)$, $\mathbf{b} = (3, -2)$, $\mathbf{c} = (1, 1)$ and $\mathbf{d} = (-1, 0)$.

(a) Compute $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$, and say which pair (if any) is orthogonal.

(b) Find the angle between $\mathbf{c}$ and $\mathbf{d}$.

Exercise 1.5 *(⋆ drill — projection).* Project $\mathbf{u} = (6, 2)$ (a) onto $\mathbf{v} = (3, 0)$ and (b) onto $\mathbf{w} = (0, 5)$.

Exercise 1.6 *(⋆ drill — cross product and area).* (a) Compute $\mathbf{i} \times \mathbf{j}$.

(b) For $\mathbf{u} = (2, -2, 1)$ and $\mathbf{v} = (1, 2, 2)$ compute $\mathbf{u} \times \mathbf{v}$ and check that it is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

(c) Hence find the area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$, and of the triangle with these as two sides.

Exercise 1.7 *(⋆ drill — equation of a line).* (a) Give the vector, parametric and symmetric equations of the line through $P(1, 2, 3)$ in the direction $\mathbf{d} = (2, -1, 4)$.

(b) Give the vector equation and the slope of the line through $A(0, 1)$ and $B(4, 3)$ in $\mathbb{R}^2$.

Exercise 1.8 *(⋆ drill — equation of a plane).* (a) Find the Cartesian equation of the plane with normal $\mathbf{n} = (1, 2, -1)$ through the point $(2, -1, 3)$.

(b) Write down a normal vector to the plane $3x - y + 2z = 7$.

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core — linear combinations)*. Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.

(a) Express $\mathbf{c} = (7, 0)$ as $\alpha\mathbf{a} + \beta\mathbf{b}$.

(b) Are $\mathbf{a}$ and $\mathbf{b}$ linearly independent, and do they form a basis of $\mathbb{R}^2$? Explain geometrically.

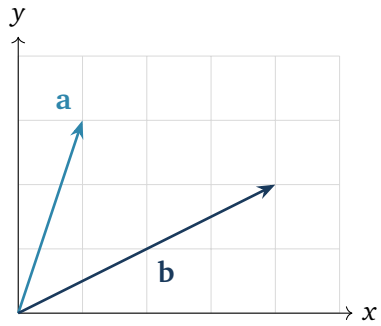
Exercise 2.2 *(▶ ⋆⋆ core, data science — angle and cosine similarity)*. Over the four-word vocabulary (data, model, train, plot) two short documents have word-count vectors $\mathbf{u} = (2, 1, 1, 0)$ and $\mathbf{v} = (1, 0, 1, 0)$.

(a) Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$ using the dot product.

(b) State their cosine similarity $\cos\theta$ (exactly, and to three decimal places) and interpret it: on the scale from 0 (no words in common) to 1 (the same word proportions), how similar are the documents?

Exercise 2.3 *(⋆⋆ core, conceptual — spot the error)*. A student writes: "For $\mathbf{u} = (1, 2, 2)$ and $\mathbf{v} = (2, 2, 1)$ we have $\mathbf{u} \cdot \mathbf{v} = 2 + 4 + 2 = 8$, so $\cos\theta = 8$ and $\theta = \arccos 8$." Identify the mistake and give the correct angle.

Exercise 2.4 *(▶ ⋆⋆ core — reading a diagram: parallel and perpendicular parts)*. The figure shows two vectors $\mathbf{a}$ and $\mathbf{b}$ drawn from the origin on a unit grid.



(a) Read off the components of $\mathbf{a}$ and $\mathbf{b}$ from the grid.

(b) Resolve $\mathbf{a}$ into a component $\mathbf{a}_{\parallel}$ parallel to $\mathbf{b}$ and a component $\mathbf{a}_{\perp}$ perpendicular to it. Verify that the two pieces are orthogonal and sum to $\mathbf{a}$.

Exercise 2.5 *(⋆⋆ core — a perpendicular direction)*. Find a vector orthogonal to both $\mathbf{a} = (1, 2, 1)$ and $\mathbf{b} = (2, -1, 3)$, and then a *unit* vector in that direction.

Exercise 2.6 *(⋆⋆ core — area of a triangle in space)*. Find the area of the triangle with vertices $A(1, 1, 0)$, $B(2, 3, 1)$ and $C(0, 2, 2)$.

Exercise 2.7 *(▶ ⋆⋆ core, physics — torque and work)*. A force $\mathbf{F} = (2, 3, 1)$ (newtons) acts at the point with position vector $\mathbf{r} = (1, 0, 2)$ (metres) measured from a pivot.

(a) Find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about the pivot.

(b) The point of application then undergoes the displacement $\mathbf{d} = (0, 4, 0)$ (metres). Find the work $W = \mathbf{F} \cdot \mathbf{d}$ done by the force.

Exercise 2.8 *(▶ ⋆⋆ core — a line, and points on it)*. Let ℓ be the line through $A(1, 0, 2)$ and $B(3, 4, 6)$.

(a) Give a parametric equation of ℓ, using the simplest integer direction.

(b) Does $C(0, -2, 0)$ lie on ℓ?

(c) Does $D(2, 1, 4)$ lie on ℓ?

Exercise 2.9 *(► ⋆⋆ core — the plane through three points).* Let $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$.

(a) Find the equation of the plane through A, B, C in point-normal and in Cartesian form, and verify that all three points satisfy it.

(b) Write the same plane in parametric form $\mathbf{r} = \overrightarrow{OA} + s\,\overrightarrow{AB} + t\,\overrightarrow{AC}$. Evaluate the point at $s = t = 1$ and check that it satisfies your Cartesian equation.

Exercise 2.10 *(► ⋆⋆ core — where a line meets a plane).* Use the plane $3x+2y+z = 6$ of Exercise 2.9.

(a) Where does the line $\mathbf{r} = t(1, 1, 1)$ meet the plane?

(b) Does the line $\mathbf{r} = t(1, -1, -1)$ meet the plane? Explain your answer using the dot product of the line's direction with the plane's normal.

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core, data science — ranking by cosine similarity).* A query is the vector $\mathbf{q} = (1, 1, 1)$. Three documents are $\mathbf{d}_1 = (2, 2, 2)$, $\mathbf{d}_2 = (1, 1, 0)$, $\mathbf{d}_3 = (0, 0, 5)$. Rank them by cosine similarity to $\mathbf{q}$, and comment on $\mathbf{d}_1$.

Exercise 3.2 *(⋆⋆ core — unit normal and equation of a plane).* Let $P(1, 0, 1)$, $Q(0, 2, 3)$ and $R(2, 0, 0)$.

(a) Find a unit normal to the plane through P, Q, R.

(b) Find the Cartesian equation of the plane.

Exercise 3.3 *(⋆⋆ core — angle between two planes).* Find the acute angle between the planes $x + y + z = 1$ and $x + z = 0$.

Exercise 3.4 *(⋆⋆ core — foot and distance from a point to a plane).* Let $P(3, 1, 2)$ and the plane $2x - 2y + z = 4$.

(a) Find the foot F of the perpendicular from P to the plane.

(b) Hence find the distance from P to the plane.

Exercise 3.5 *(⋆⋆ core, physics — work done sliding down a ramp).* A crate of mass 10 kg slides down a ramp, undergoing the displacement $\mathbf{s} = (6, -8)$ (metres), with the $+y$ axis pointing up. Taking $g = 10\,\mathrm{m/s^2}$, the weight is the force $\mathbf{F} = (0, -100)$ (newtons).

(a) Find the work $W = \mathbf{F} \cdot \mathbf{s}$ done by gravity.

(b) Find the *scalar* component $\mathbf{F} \cdot \hat{\mathbf{s}}$ of the weight in the direction of motion.

Exercise 3.6 *(⋆⋆ core — expansion identities, the parallelogram law and the rhombus).* Using only $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$, the distributive rule $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$, the scalar rule $(\lambda \mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v})$ and commutativity $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$:

(a) show that $\|\mathbf{u} \pm \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2\,\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2$;

(b) deduce the parallelogram law $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\,\|\mathbf{u}\|^2 + 2\,\|\mathbf{v}\|^2$, and say what it means for the diagonals and sides of a parallelogram;

(c) a rhombus has sides $\mathbf{u}, \mathbf{v}$ with $\|\mathbf{u}\| = \|\mathbf{v}\|$ and diagonals $\mathbf{u} \pm \mathbf{v}$; prove that its diagonals are perpendicular.

Exercise 3.7 *(⋆ ⋆ ⋆ challenge, optional enrichment — coplanarity of four points).* Let $P(0, 0, 0)$, $Q(1, 1, 0)$, $R(1, 0, 1)$.

(a) Decide whether P, Q, R and $S(0, 1, 1)$ are coplanar. If not, give the volume of the parallelepiped on $\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$.

(b) Decide whether P, Q, R and $T(2, 1, 1)$ are coplanar.

Exercise 3.8 *(⋆⋆ core — foot of the perpendicular to a line).* Let ℓ be the line through the origin in the direction $\mathbf{d} = (1, 2)$, and let $P(5, 0)$. Find the foot F of the perpendicular from P to ℓ, and the distance from P to ℓ.